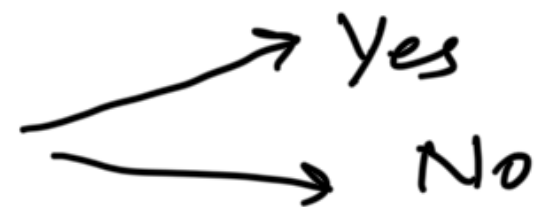


Decision Problems


 $f: \mathbb{N} \rightarrow \{0, 1\}$ under an appropriate encoding

Decision vs Optimization ($g: \mathbb{N} \rightarrow \mathbb{N}$)

Decision-Knapsack : Given weights w_i
 profits p_i , capacity B
 and a number $m \in [1 \dots n \cdot p_{\max}]$

Is the optimal solution $O^* \geq m$?

Claim : (i) If we can solve the optimization problem efficiently, we can solve the decision version efficiently

(ii) If we can solve the decision version

efficiently, we can solve the optimization
efficiently use binary search in $[1 \dots n \cdot p_{max}]$

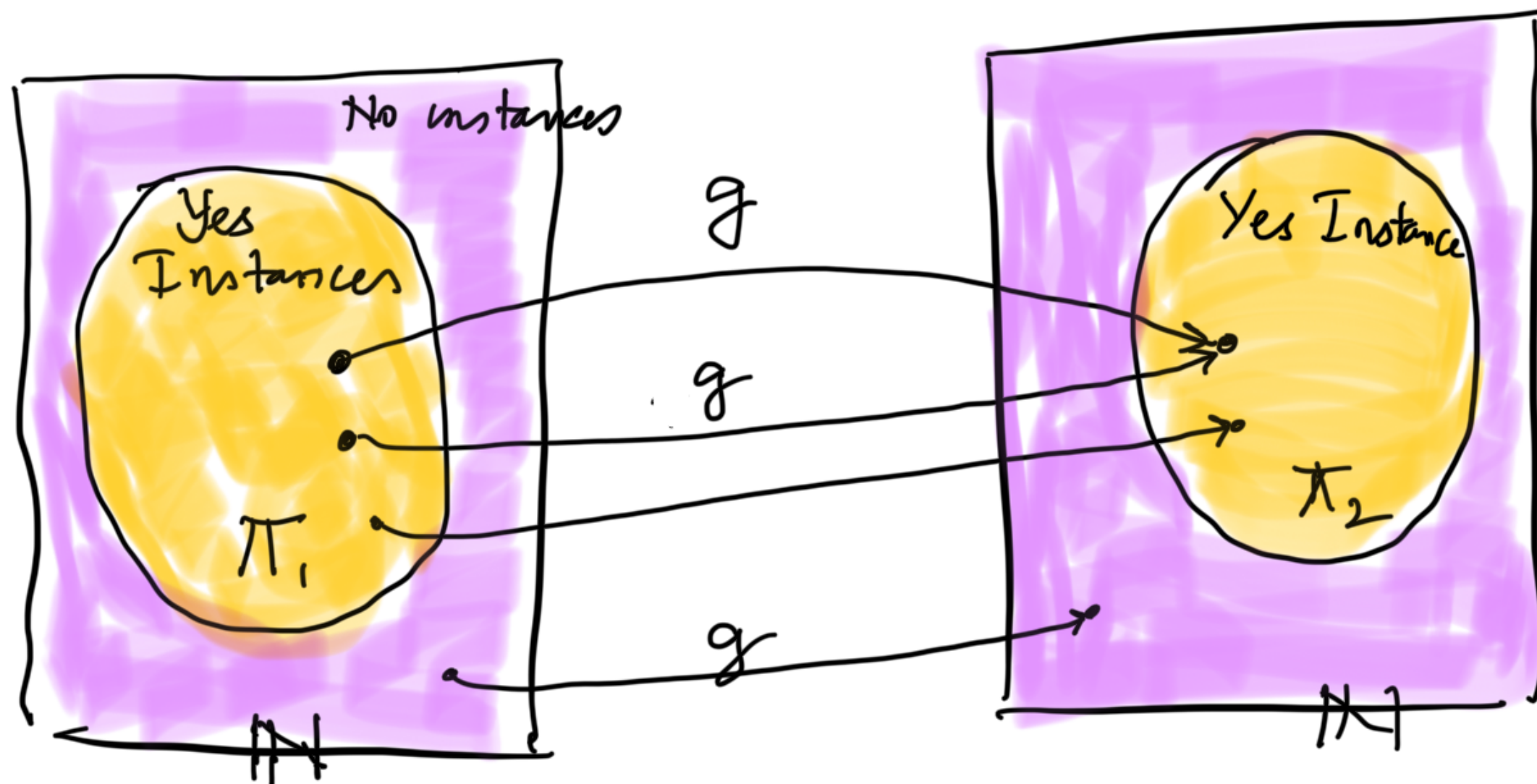
Efficiently: polynomial time $O(n^c)$ for some constant c

How do we relate computational complexity of problems

For problems A, B

- (i) A is similar to B
- (ii) A is harder than B
- (iii) B is harder than A

Reducibility (relation) : Notation α or $\leq$



π_1 is reducible to π_2
 $\pi_1 \leq_g \pi_2$
 $g : \mathbb{N} \rightarrow \mathbb{N}$ (many-one)
reduction function
 $\Leftrightarrow x \in \pi_1 \Leftrightarrow g(x) \in \pi_2$

If g is computable in $t(n)$ time / $s(n)$ space

$$\pi_1 \leq_{t(n)} \pi_2 \quad / \quad \pi_1 \leq_{s(n)} \pi_2$$

Important cases

(i) $t(n)$: polynomial time

π_1 is polynomial time reducible to π_2

(ii) $s(n)$ logarithmic space (additional)

π_1 is logspace reducible to π_2

$\leq$: partial order

Observation

$\leq^{\text{poly}}$

is transitive, i.e.

$A \leq^{\text{poly}} B$ and $B \leq^{\text{poly}} C$, then $A \leq^{\text{poly}} C$

Proof There exist q_1, q_2 s.t. $A \leq^{\text{poly}} B$ $B \leq^{\text{poly}} C$

and g_1, g_2 are polynomial-time

Consider $g_2(g_1)$; then $A \leq g_2(g_1) \subseteq$

Is $g_2(g_1)$ polynomial time computable?

Claim

Let $\pi_1 \leq^{\text{poly}} \pi_2$ then

(i) If π_2 can be computed in polynomial time, then π_1 can be computed in poly-time

(ii) If there is no polynomial time algorithm for π_1 , then there cannot be any polynomial time algorithm for π_2

P : the class of decision problems
which ^{can} have polynomial time algorithms.

NP: the class of decision problems for which
Non-deterministic₂ we have polynomial time verification
Polynomial

Clearly $P \subseteq NP$
 $P \stackrel{?}{=} NP$

(but is the
containment strict)

π is NP-hard if

$\forall A \in \text{NP}, A \leq^{\text{poly}} \pi$

π is NP complete (i) π is NP-hard
(ii) $\pi \in \text{NP}$

Is there any NPC Problem?

Cook-Levin Thm : Satisfiability of Boolean formula
is NPC